

● RAG Learning

B5.2 Natural Selection & Evolution

OCR A GCSE Biology (J247) Gateway — Topic B5: Genes, inheritance and selection

1. Variation

Variation = differences between individuals of the same species. Variation is essential for evolution.

Source	Examples
Genetic	Eye colour, blood group, attached/free earlobes — passed down from parents
Environmental	Tan from sun, scars, language spoken
Both	Most traits — height (genes + nutrition), weight (genes + diet/exercise)

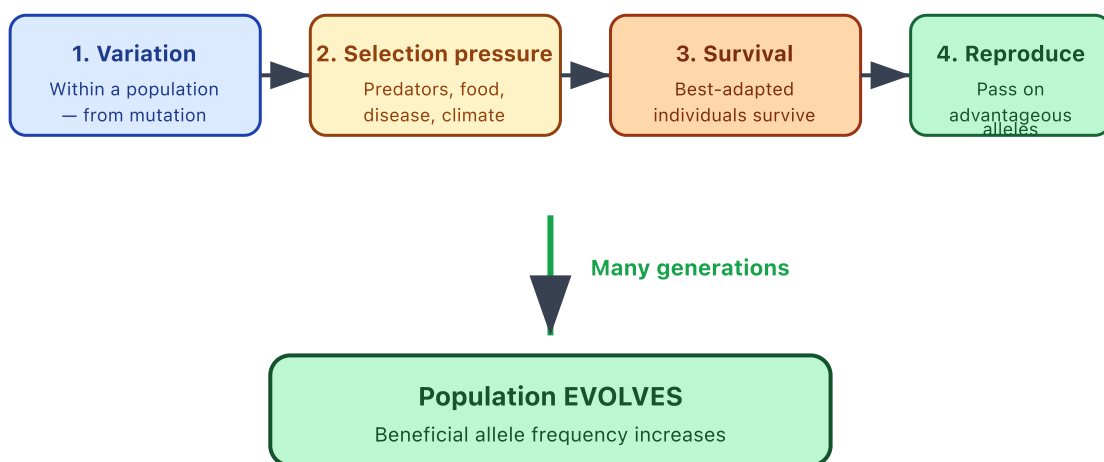
Continuous vs Discontinuous variation

	Continuous	Discontinuous
Type	Range of values (any number)	Distinct categories (no in-between)
Examples	Height, mass, leaf length	Blood group (A, B, AB, O), gender, eye colour categories
Caused by	Many genes + environment	One or few genes
Graph	Histogram / bell curve	Bar chart with separate bars

2. Evolution by Natural Selection

Evolution = a change in the inherited characteristics of a population over many generations through natural selection. Proposed by **Charles Darwin** in 1859 in *On the Origin of Species*, based on his observations during the voyage of HMS Beagle (especially in the Galápagos Islands).

Natural Selection — Four Key Steps



The four steps in detail

1. **Variation** exists within a population (caused by mutation, sexual reproduction).
2. A **selection pressure** acts on the population (e.g. predator, food shortage, climate change).
3. Individuals with characteristics best suited to the pressure (most "fit") **survive longer** and **reproduce more**.
4. The advantageous alleles are **passed to the next generation**. Over many generations, the population becomes increasingly adapted — **evolution** has occurred.

3. Evidence for Evolution

Evidence	What it shows
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Fossils	Organisms preserved in rocks reveal how species have changed over time. Sequence of horse fossils shows gradual increase in size and reduction in number of toes.
Antibiotic-resistant bacteria	Resistance evolved within decades of antibiotic use — direct observation of natural selection in action.
Comparative anatomy	Pentadactyl limbs (5 digits) in mammals — same basic structure for different uses (running, swimming, flying) suggests common ancestor.
DNA sequencing	Closely-related species have very similar DNA. The more similar, the more recent the common ancestor.

Why is the fossil record incomplete?

- Most organisms decay before fossilising.
- Soft-bodied organisms (worms, jellyfish) rarely fossilise.
- Many fossils have been destroyed by geological activity (tectonics, erosion).
- Many fossils have not yet been discovered.

4. Antibiotic Resistance — Evolution in Action

Bacteria reproduce rapidly. Random **mutations** sometimes give resistance to antibiotics. When antibiotics are used, only resistant bacteria survive and multiply — leading to a population dominated by resistant bacteria. **MRSA** is a famous "superbug" with resistance to many antibiotics.

How resistance spreads

1. Random mutation in one bacterium gives resistance to an antibiotic.
2. Antibiotic kills non-resistant bacteria; resistant ones survive.
3. Resistant bacterium reproduces rapidly (binary fission); population becomes increasingly resistant.

4. Resistance can spread between bacteria via **plasmids** (horizontal gene transfer).

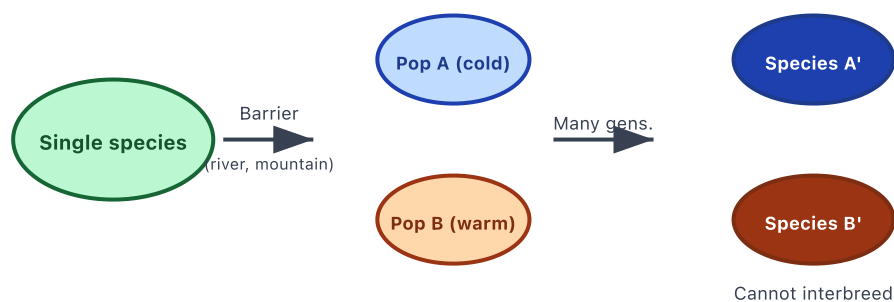
Reducing the development of resistance

- Doctors should not prescribe antibiotics for non-bacterial illnesses (e.g. viral colds).
- Patients must finish the full course (so all bacteria are killed, none survive to develop resistance).
- Restrict use of antibiotics in farming.
- Develop new antibiotics — but slow and expensive.

5. Speciation

Speciation is the formation of a new species. It typically happens when two populations of the same species become **isolated** (often geographically), experience **different selection pressures**, and gradually become so genetically different that they can no longer interbreed to produce fertile offspring.

Speciation by Geographical Isolation



Steps to speciation

1. Population becomes isolated (geographically).
2. Different environmental conditions = different selection pressures on each isolated group.
3. Mutations + natural selection cause each population to adapt to its environment.
4. After many generations, the populations become so genetically different that they can no longer interbreed to produce fertile offspring → they are now **different species**.

6. Selective Breeding (Artificial Selection)

Selective breeding is when humans choose individuals with desired characteristics and breed them together over many generations. Examples: high-yield wheat, disease-resistant crops, dairy cows producing more milk, dogs bred for appearance or behaviour.

Steps

1. Choose parents with the desired characteristic from a varied population.
2. Breed them together.
3. From the offspring, select those with the most desired characteristic.
4. Breed those individuals together. Repeat over many generations.

Issues: reduced gene pool / inbreeding (more recessive genetic disorders); some breeds suffer health problems (e.g. bulldogs with breathing difficulties).

7. Classification

Classification is the system of grouping organisms based on their characteristics. The system used today is largely based on Carl Linnaeus's **binomial naming system** — every species has a two-part Latin name (e.g. *Homo sapiens*, *Panthera leo*).

The hierarchy

Kingdom → Phylum → Class → Order → Family → Genus → Species

Mnemonic: *King Philip Came Over From Greece Singing*.

The Three Domains (proposed by Carl Woese)

Domain	Examples
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Archaea	Single-celled prokaryotes living in extreme environments (hot springs, deep ocean vents)
Bacteria	True bacteria (most familiar prokaryotes)
Eukarya	All organisms with eukaryotic cells: animals, plants, fungi, protists

Evolutionary trees

An **evolutionary tree** shows the relationships between species and their common ancestors. Built using fossil records, anatomical comparisons, and DNA sequence similarity.